PRANAY PANDEY

2K23/CS/308 – DTU

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EXPERIENCE

Product (SWE) Intern, Adobe May 2026–Jul 2026

Technologies: Python, Scala, AWS-Bedrock Anthropic Models, Azure OpenAI Services, Langfuse, Jenkins CI/CD, Git, Jira

- Architected a GEPA-inspired, **feedback-based prompt optimization loop** for the MAS LIPI agent, integrating deterministic evaluation gates and LLM-driven candidate generation, elevating the system pass rate from **76% to 87%** across 628 test scenarios with **zero runtime overhead**.
- Orchestrated rigorous LLM telemetry via Langfuse and Jenkins CI/CD pipelines. Benchmarked GPT-5.2 against 5.4, 5.4-mini (low/med/high reasoning), 5.4-nano, GPT-5-mini, o3, 4o-mini, and o4-mini. Proved GPT-5.4-mini (low reasoning) **achieved a 57.5% operational cost reduction** over the **2.35x more expensive** GPT-5.2 baseline while sustaining a 97.0% deterministic-set pass rate.
- Engineered **Meta Harness Skill-Centric (MHSC), an autonomous, agent-agnostic, multi-surface harness optimization framework** inspired by Stanford's Meta-Harness and the Ralph Loop; evolved its Claude-Skill architecture across Adobe's MAS stack (Halo, LIPI, Rules) to automate prompting guide-driven edits, iterative benchmarking, and **reducing ~3 weeks of manual LLM model migration to an overnight run**.
- Deployed MHSC as a zero-overhead execution pipeline **requiring only target endpoints and API keys** to perform agent discovery, self-evaluation, and automated PR generation. MHSC Early Wins reduced per-scenario token consumption by **12.33% in LIPI** and **27.0% in Halo**.

Machine Learning Intern, Defence Research and Development Organisation (DRDO) Jun 2025–Jul 2025

Technologies: Python, Pandas, NumPy, NLTK, Scikit-learn, Node2Vec, K-Means, Fuzzy C-Means, FuzzyWuzzy, FastAPI, SQLite, Tortoise ORM

- Created a high-throughput **Python Microservice** for entity recommendation using FastAPI, SQLite, and Tortoise ORM; delivered standards-driven RESTful APIs across a microservice architecture with 100% deployment success and zero production rollbacks.
- Architected a **hybrid recommendation engine** combining K-Means, Fuzzy C-Means, Node2Vec graph embeddings, and TF-IDF-enhanced fuzzy matching—achieving sub-second NLP retrieval with a **530ms average response time** in high-volume environments.

EDUCATION

B.Tech (Computer Science Engineering)	2023–2027	Delhi Technological University, New Delhi	7.765 CGPA
CBSE (Class XII)	2021–2023	Sant Gyaneshwar Model School, Delhi	83.6%
CBSE (Class X)	2009–2021	Hansraj Model School, Punjabi Bagh	94%

ACADEMIC PROJECTS

RAG-Driven Document Q&A and Smart Chat-Bot Platform (GitHub Link)

Technologies: Python, LlamaIndex, BM25, CrossEncoders, Groq API (LLaMA-4), Hugging Face, FastAPI, Node.js, Express, MongoDB

- Architected a Python RAG microservice, implementing a **2-stage hybrid retrieval pipeline (Dense + BM25)** fused via Reciprocal Rank Fusion (RRF). Integrated a HuggingFace CrossEncoder to jointly score query-passage pairs, surfacing the top-3 most **contextually relevant** chunks.
- Accelerated throughput by developing a cosine-similarity **semantic caching** layer over query embeddings, reducing response latency by **98.3% (from 3.55s to 60ms)**. Designed a deterministic 13-class intent router to bypass generative LLM inference for static interactions.

Modular C++ Ride-Sharing System — Spatial Matching Engine (GitHub Link)

Technologies: C++, Drogon Web Framework, Domain-Driven Design (DDD), RESTful APIs, Atomics, Object Pool, Spatial Grid

- Architected a C++ backend utilizing 4-layer DDD with decoupled pricing and matching engines. Engineered a custom freelist Object Pool over contiguous `aligned_storage_t`, achieving **27ns** allocation latency by bypassing kernel overhead for **1,000+** GPS telemetry events/sec.
- Developed a high-performance Drogon REST API featuring a cache-friendly Spatial Grid that replaced linear $O(n)$ scans with $O(1)$ lookups, achieving **26μs matching latency (a 17.4x speedup)** for 10,000 drivers. Integrated lock-free surge pricing, ensuring zero heap allocations.

YOLO-Powered Real-Time Multi-Class Object Detection System (GitHub Link)

Technologies: OpenCV, PyTorch, NumPy, Pandas, Scikit-learn, Ultralytics-YOLOv8, COCO-Dataset, Convolutions, Transfer Learning

- Developed and fine-tuned an efficient YOLOv8-nano detection system using COCO pre-trained weights and custom traffic annotations, achieving **85%+ accuracy** and **24+ FPS** real-time performance on live video streams using standard CPU hardware.
- Implemented **automated** data pre-processing workflows resulting in **72% data quality improvement**, and deployed multi-platform inference pipelines optimized for edge computing **(2GB deployment)** and cloud scalability via containerized deployment.

TECHNICAL SKILLS

- **Core Programming Languages:** Python, Java, C/C++, Rust
- **ML & AI:** Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, OpenCV, YOLOv8, Transformers, Hugging Face, LlamaIndex, Generative Models
- **Databases & Infrastructure Tools:** Redis, MongoDB, MySQL, AWS S3, Neo4j, Git, Docker, Linux, Bash Scripts
- **Computer Science Fundamentals:** System Design (HLD, LLD), Operating Systems, Computer Architecture & Organization, Compiler Design, Distributed Systems, Parallel Computing, Computer Networks, Database Management, Object-Oriented Programming
- **Web Development:** HTML, CSS, Bootstrap, JavaScript, React, Express, Node.js, FastAPI, REST APIs, GraphQL, NextJS, Vercel, OAuth2

ACHIEVEMENTS AND CERTIFICATIONS

- **2nd Place, Adobe Invictus DevCraft Hackathon** – Programmed a high-precision Naive Bayes Document Classifier (95%+ accuracy)
- **Machine Learning Specialization** (DeepLearning.AI & Stanford University) – Supervised Machine Learning, Unsupervised Learning, Reinforcement Learning and Advanced Learning Algorithms (Regression, Clustering, Ensembles and Recommender Systems)– (Certificate Link)
- **Deep Learning Specialization** (DeepLearning.AI) – Neural Networks and Deep Learning, Sequence Models, Convolutional Neural Networks (Natural Language Processing, Transfer Learning, Transformers, RNNs, LSTMs and Hugging Face) – Certificate Links: (CL1) (CL2) (CL3)
- **Generative Adversarial Networks (GANs) Specialization** (DeepLearning.AI) – (Certificate Link)
- **250+ LeetCode** Data Structures and Algorithms Problems Solved (Python and Java) – (LeetCode Link)